

# Detecting Phishing Websites with Explainable Machine Learning

Feature Engineering, Model Comparison, and Interpretable Risk Assessment

## Abstract

Phishing remains a major cybersecurity problem because attackers continuously modify domains, page layouts, and delivery mechanisms. This paper proposes an explainable machine-learning framework for classifying websites as legitimate or phishing. The approach combines URL-based, host-based, and page-structure features with interpretable classifiers. Several model families are compared conceptually, and feature attribution is used to explain individual predictions. The paper argues that high predictive performance is insufficient for security operations unless analysts can understand why a URL was classified as risky. An evaluation framework is presented that emphasizes precision, recall, F1 score, false-negative rate, and interpretability.

**Keywords:** phishing detection; machine learning; explainable AI; cybersecurity; URL analysis

## 1. Introduction

Phishing attacks exploit human trust by presenting fraudulent websites that imitate legitimate services. Traditional blocklists are effective against known threats but can struggle with newly registered domains and rapidly changing campaigns. Machine learning offers an opportunity to identify patterns associated with malicious websites before they are explicitly catalogued.

The central challenge is balancing detection capability with operational interpretability. A security analyst needs more than a binary label; the analyst may also need evidence indicating that a URL contains suspicious characteristics. This paper therefore focuses on explainable classification rather than accuracy alone.

## 2. Feature Representation

The framework represents each website using multiple feature groups. URL features include length, number of subdomains, unusual characters, token patterns, and the use of an IP address instead of a conventional domain. Host-level features can include domain age and certificate-related signals when such information is available. Page-level features may describe forms, external resources, redirects, and suspicious scripting behavior.

Combining heterogeneous features can improve robustness because an attacker may manipulate one feature group while leaving others unchanged. However, features that require external network lookups introduce latency and availability concerns.

## 3. Experimental Design

A balanced experimental dataset should contain both legitimate and phishing examples collected from independent sources and separated chronologically when possible. Chronological evaluation is important because random splits can overestimate generalization when similar campaigns appear in both training and test sets.

The proposed comparison includes logistic regression, decision trees, random forests, and gradient-boosted models. Accuracy is reported alongside precision, recall, macro F1, and false-negative rate. Explainability is evaluated through global feature importance and local attribution for individual predictions.

## 4. Discussion

Tree-based ensembles are expected to perform strongly because phishing indicators often involve nonlinear interactions. For example, a long URL may not be suspicious by itself, but a long URL combined with unusual

subdomain structure and an IP-based host can substantially increase risk. Logistic regression remains valuable as a transparent baseline.

Explainability can reveal both useful signals and potential weaknesses. If a model relies excessively on a single unstable feature, an attacker may be able to evade detection by manipulating that feature. Security-oriented evaluation should therefore include adversarial and temporal testing rather than relying solely on an IID test set.

## 5. Conclusion

An effective phishing detector should be evaluated as a decision-support system rather than merely a classifier. Combining multiple feature groups with interpretable machine-learning techniques can provide useful detection while giving analysts evidence for investigation. Future research should examine concept drift, adversarial manipulation, privacy-preserving feature collection, and calibration of predicted risk scores.

## References

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